Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/09/2005
PI Name:	DeTitta, George T. Ph.D.		
Project Title:	High Throughput Methods for Rational Prediction of Macromolecular Crystallization Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Buffalo	**State:**	NY
Zip Code:	14203-1196	**Congressional District:**	28
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	07/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Luft, J. R. (Hauptman-Woodward Medical Research Institute)		
Grant/Contract No.:	NAG8-1839		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A ground-based program is underway to test the hypothesis that similar proteins will crystallize under similar conditions. The first phase of the project is the execution and evaluation of the outcomes of precipitation reactions. Hundreds of macromolecular solutions will be individually incubated with 1536 reagents that have the potential to reduce their solubility. It is our hypothesis that the pattern of outcomes in these reactions will provide an objective measure of similarity. The second phase of the project will be undertaken in the physical chemistry laboratory. We are measuring ancillary physical chemical information for hundreds of proteins (sodium dodecyl-sulfate polyacrylamide gel electrophoresis, Dynamic Light Scattering) and thousands of chemical reagents (refractive index, pH, viscosity, vapor pressure, surface tension). We will mine this data to look for correlations that may be predictive for crystallization. The third phase of the project is being undertaken in the computer laboratory. Images of the precipitation reactions are being scored to produce a unique code. This index will represent the outcome of the precipitation reactions of an individual protein. The final stage of the project will test our hypothesis. We will compare the crystallization properties of protein		

Task Description:	pairs with similar precipitation reaction indices. If the crystallization properties and precipitation reaction indices are similar we should be able to rationally predict crystallization conditions for a protein based upon the index. Our objective is to predict crystallization conditions for macromolecules that have not previously been crystallized. The work will focus on the development of an objective measure of similarity between macromolecules. The measure of similarity will be used as a rational, predictive tool for macromolecular crystallization. The detailed, three-dimensional view of macromolecular structure that is provided by X-ray crystallography produces structural information required for modern biology. Formation of a suitable crystal is a primary step in X-ray crystal structure determination. The process relies on random screening of chemical and physical conditions until a set of parameters that produces a crystal can be identified. Rationalizing the identification of these conditions would benefit any discipline that utilizes X-ray crystal structures in their research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A crystallization laboratory designed to determine initial crystallization conditions for biological macromolecules was established at HWI. The laboratory was made highly automated through in-house hardware and software modification of commercially available robotic systems. This process has continued over the past year. The capabilities of the laboratory database are enhanced. Queries of the data to look for trends in the nearly 5000 macromolecular samples (7.5 million crystallization experiments) set up from February 2000 – February 2005 are underway. We are currently investigating the chemical components of the macromolecular samples that were screened in the laboratory. As many as 200 unique macromolecular samples are used to generate 300,000 crystallization experiments each month. During that same month 1.8 million images of these experiments' outcomes are recorded. An active community of more than 450 structural biologists and two structural genomics centers provide macromolecular samples to the facility. The database and computer infrastructure automate the majority of data transfer (images of the experiment outcomes) to researchers through a secure ftp server. Outcomes from a group of samples provided by the structural biology community and set up in the laboratory over a recent 12 month period were analyzed; 60% of these samples provided one or more conditions for follow-up crystallization trials. Determining initial crystallization conditions is a critical and primary step in the X-ray structure determination process. The samples provided by the structural biology community represent a diverse collection of biologically important macromolecules. The three dimensional structures of these macromolecules are used by scientists to better understand molecular structure and function. This information can lead to the identification of macromolecules that are potential drug targets and in some cases will lead to the development of new drugs and treatments to improve human health. The screening process continually evolves and benefits from the large numbers of diverse biological macromolecules provided by the structural biology community. The 1536 crystallization cocktails have been refined on an annual basis to decrease the number of false leads. New 1536 well plates are in production by Greiner Bio-One. The plates were specifically prototyped for and tested in the screening laboratory. The plates are constructed with a low-birefringence plastic. This property of the new plate allows imaging with a cross-polarized light source. Crystals located outside of the plane of focus, or polycrystalline material that may appear similar to a phase separation or precipitate under standard lighting are clearly distinguishable as crystalline material under cross-polarization. We have refined a robotic method to add nanoliter quantities of dye to experiment plates. The dye can aid in the identification of macromolecular crystals. Color imaging methods are under development to archive the experimental outcomes of the dyed plates. Additional failover systems in the laboratory's computer infrastructure as well as the construction of a third imaging robot contribute to the capacity and stability of the laboratory's throughput. Our library of manually scored, "truth data" (for use in developing methods of automated image analysis) now exceeds 1.5 million images. Image data has been sorted into the two categories 1) something worth pursuing with follow-up crystallization experiments and 2) nothing worth pursuing. Software for automated image analysis is under development at the Hauptman-Woodward Institute and with groups at the University of Toronto and at Columbia University.
Bibliography Type:	Description: (Last Updated: 03/26/2008)
Presentation	Luft, J.R., Collins, R.J., Fehrman, N.A., Gulde, S.M., Lauricella, A.M., Mancuso, C.A., Li, J., Carroll, B., Fan, E., Robien, M., Veatch, C.K., Hol, W.G.J. and DeTitta, G.T. "Co-crystallant screening in Buffalo" Annual Structural Genomics of Pathogenic Protozoa Meeting at the University of Washington, Seattle, WA Mar-2004
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